Assignment 7
Due at 11:59 pm on Wednesday, November 11
We will deduct 1 point from each question/subsection that is not properly assigned in Gradescope.

Reading: Sections 12.1 - 12.8 and 14.1- 14.2 of the textbook.

1. (20 Points) Is a balanced outcome in a network exchange experiment (where participants are trying to maximize their payoff) always stable? If it is, prove it. If not, give an example of a balanced outcome that is not stable.
   Note: You're allowed to use the result from this question in the rest of the assignment.

2. (18 Points) Consider a network exchange experiment on the following network, where the thick lines represent edges along which an exchange occurs.

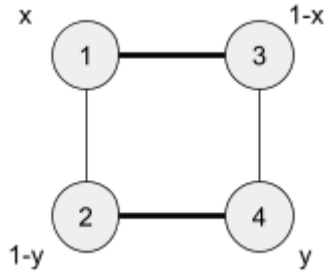
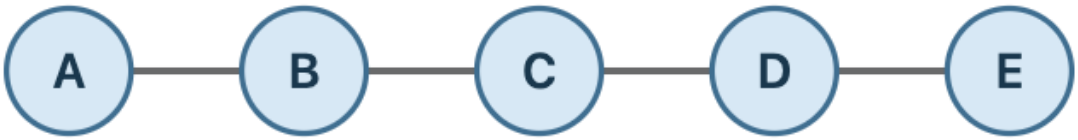


Figure 1.

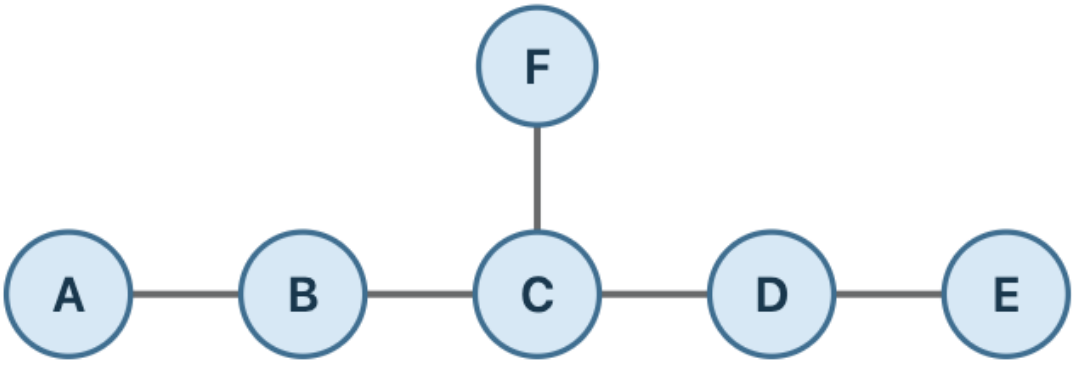
   a. If x = y = 0.3 in Figure 1, is the outcome stable? Is it balanced? Explain.
   b. If x = 0.3 and y = 0.4, is the outcome stable? Is it balanced? Explain.
   c. Give an outcome of the network exchange in Figure 1 that simultaneously satisfies the following two conditions: (i) the outcome is balanced; and (ii) adding one new non-exchanging edge to the network would make the outcome unbalanced. Explain why the outcome satisfies both conditions.
   d. Give an outcome of the network exchange in Figure 1 that simultaneously satisfies the following two conditions: (i) the outcome is balanced; and (ii) there's no way to add a new non-exchanging edge to make it unbalanced. Explain why the outcome satisfies both conditions.
      An equivalent way of explaining the conditions in part d: (i) the outcome is balanced; and (ii) no matter what new non-exchanging edge you add to the network, the network remains balanced.

      Note: For parts c and d, you need to consider what would happen if you add a new non-exchanging edge between two of the nodes that are <u>already</u> in the network – <u>without</u> adding any new nodes.

3. (20 Points) Consider a network exchange experiment with five participants, A, B, C, D, and E, connected as shown below. Each edge represents a possible exchange relationship. In each round, a participant can exchange with at most one neighbor. Each exchange generates a total of $10, which the two participants divide between themselves.

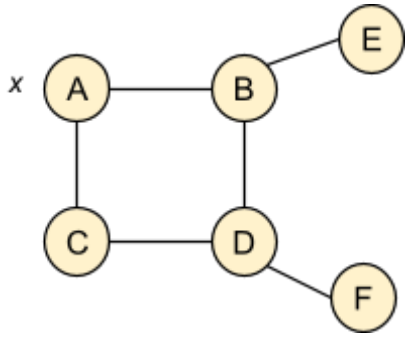


   a. Which participant or participants would you expect to receive the largest share of the money in this network? Briefly explain your reasoning. You do not need to calculate exact payoffs.
   b. Suppose that A exchanges with B and C exchanges with D, while E does not exchange. A receives a, B receives $10 - a$, C receives x, and D receives $10 - x$. Determine the best outside option for each of A, B, C, and D in terms of a and x.
   c. Using the outside options from part (b), determine the Nash bargaining solution for the exchange between A and B. Express the resulting payoffs for A and B in terms of x.
   d. Now suppose a sixth participant, F, joins the network and is connected only to C, as shown below.



   Compared with the original network in part (a), how would you expect the relative bargaining power of C and its neighbors to change? Explain your reasoning in terms of exchange opportunities and competition. You do not need to calculate exact payoffs.

4. (25 Points) Consider a Network Exchange experiment on the network below. In the outcome of this exchange, A gets $x where x can be any number between and including 0.5 and 1.



   a. If the outcome is stable, which pairs of nodes must exchange with each other? Explain in 3-5 sentences.
   Hint: First, explain why, if B doesn't exchange with E, at least one edge will have an instability. Then, use the one-exchange rule to explain which other nodes have to exchange with each other if B-E is an exchanging edge.

   b. Draw an example of a network exchange outcome that is stable *but not* balanced. Your example should specify which pairs of nodes negotiate an exchange and how much money each node receives. Show that your outcome is stable *but not* balanced.

   c. Draw an example of a network exchange outcome that is stable *and* balanced when x = 0.5. Your example should specify which pairs of nodes negotiated an exchange and how much money each node received. Show that your outcome is balanced.
   Hint: First figure out how much A and the node A is exchanging with get. Then solve the Nash bargaining equation for the edge B-E with the outside options you know. Finally solve the Nash bargaining equation for the third exchanging edge.

   d. Next, we'll walk you through some steps to specify all possible network exchange outcomes that are stable and balanced. We know that A gets $x \geq 0.5$ from their exchange. In the next 4 steps, we'll see how we can figure out what the rest of the nodes get *in terms of x*. This allows us to express all possible outcomes since for any value of *x*
      i. Who is A's exchanging partner, and how much do they get in terms of *x*?
      ii. Show that, in a stable, balanced outcome, D gets a value of *x* or more.

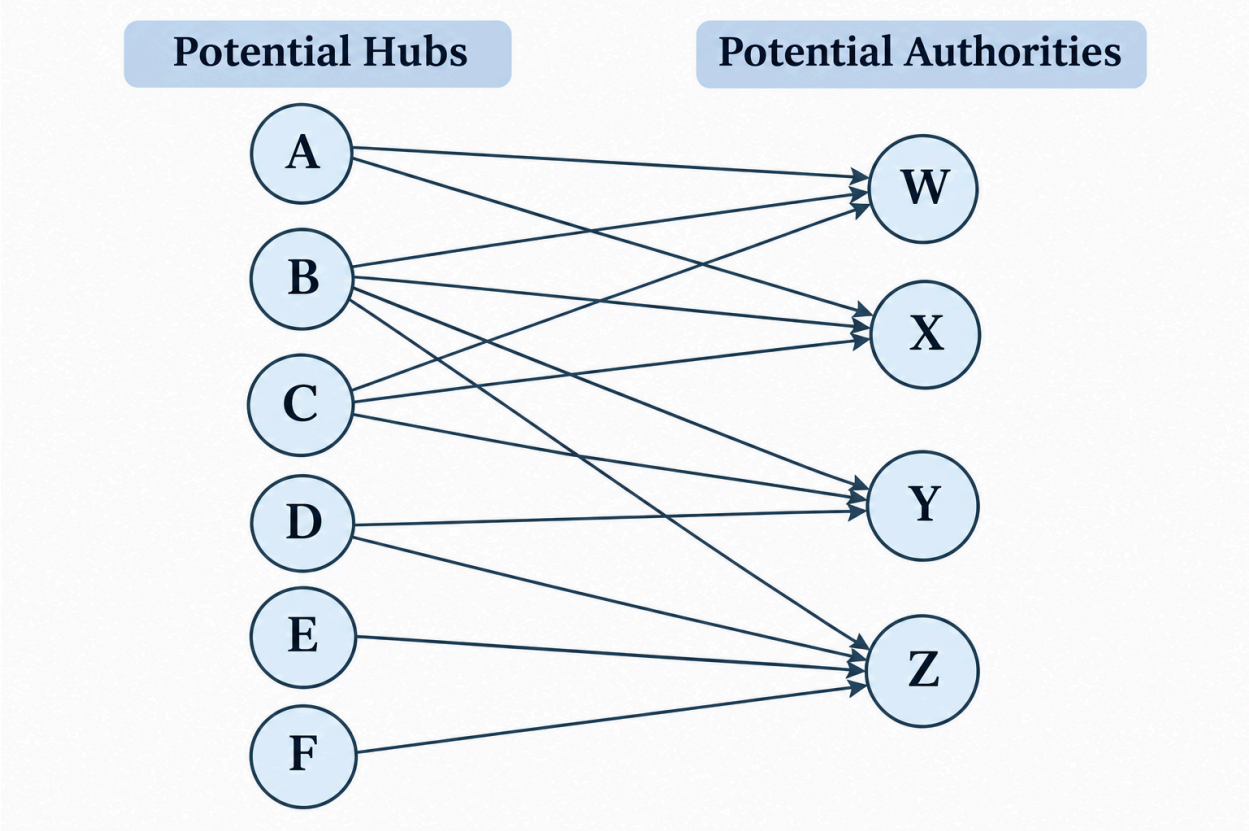
iii. In a stable, balanced outcome, how much do B and E get in terms of x? Show your work, including how you calculate each node's outside options and apply the Nash bargaining solution.

iv. There's one final exchanging pair whose split we haven't yet calculated. In a stable, balanced outcome, how much does each of them get in terms of x? Show your work, including how you calculate each node's outside options and apply the Nash bargaining solution.

e. Check your work: If you plug in $x = 0.5$ to your answers from part (d), do you get your answers from part (c)?

f. Notice that node D never gets less than node B in the outcome, suggesting that D has more power than B. What about D's position in the network makes this the case? Offer a 1-3 sentence intuitive explanation.

5. (17 Points) Consider the webpage network below. Pages A-F are potential hubs and pages W-Z are potential authorities. Start with every hub score and every authority score equal to 1.



a. Run two full HITS updates on the network. Fill in the table with the authority scores for W-Z and the hub scores for A-F after each update. Then report the normalized scores.

Scores	W	X	Y	Z	A	B	C	D	E	F
Update #1										
Update #2										
Normalized										

b. Which page has the highest normalized authority score, and which page has the highest normalized hub score after two updates? Briefly explain why each page scores highly using the HITS intuition.

c. Page A is allowed to add exactly one new outgoing link while keeping its existing links to W and X. It can add either A -> Y or A -> Z. Starting again with all hub and authority scores equal to 1, which option gives A the larger normalized hub score after two full HITS updates? Show enough calculation to justify your answer.